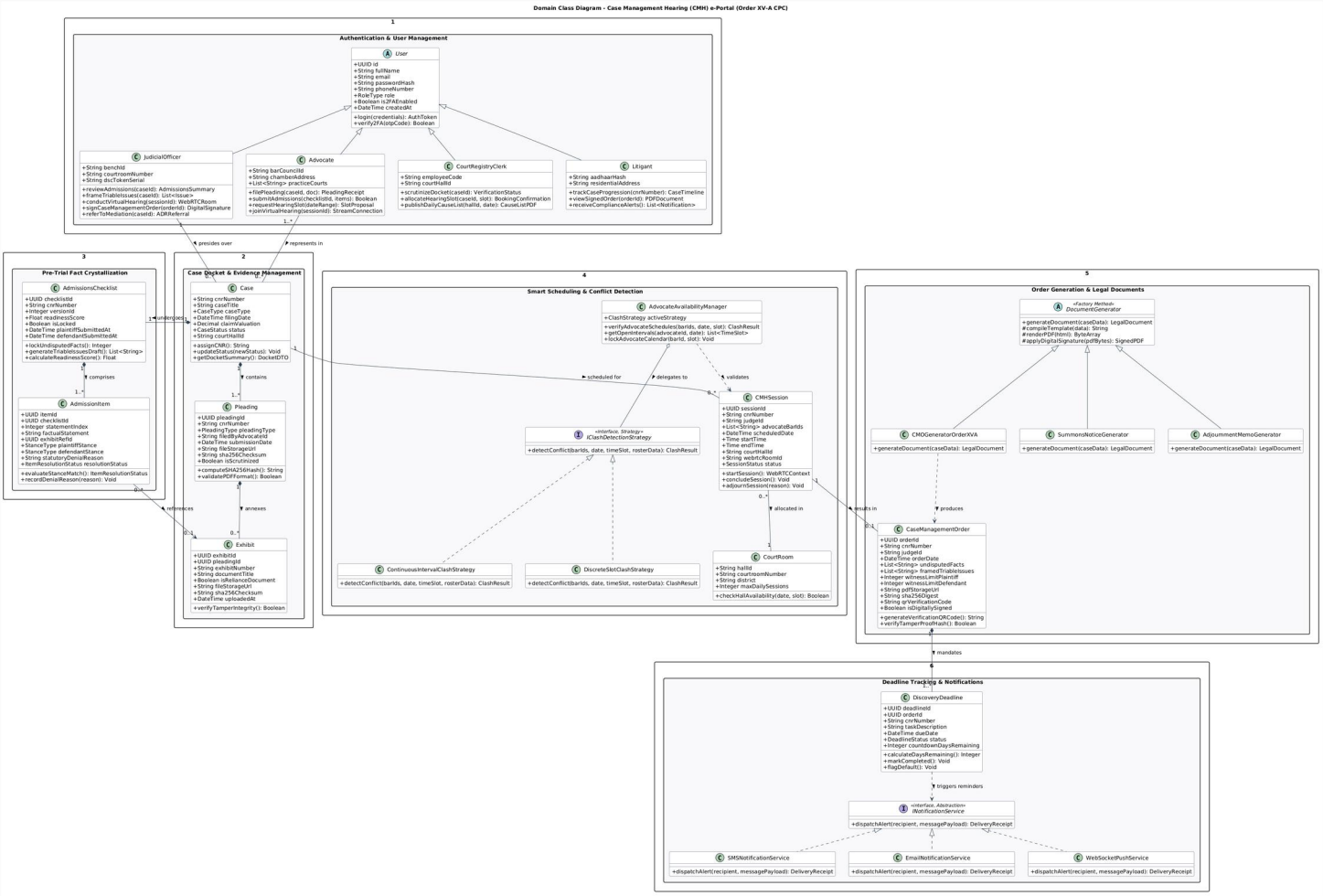
3. UML Class Diagram

Domain Object-Oriented Architecture, Encapsulation, and ORM Entities



ARCHITECTURAL & STATUTORY CPC HIGHLIGHTS:

- Role Inheritance: Abstract User base class specialized into Judge, Advocate, Clerk, and Litigant domain models.
- Data Encapsulation: Strict private attributes for password hashes and DSC keys with public verification contracts.
- Statutory Associations: Case maintains 1:N AdmissionMatrix, 1:1 CaseManagementOrder, and 1:1 EChallanPayment.
- SQLAlchemy 2.0 ORM: Maps domain entities to relational tables with foreign keys and database constraints.